Ultra Evaluation Function

How Laser War measures a position

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The central idea. A positive score favors the player whose turn it is. A negative score favors the opponent. The score combines king safety, future laser access, permanent laser assignments, route speed, and the lasers' current impact.

1. The complete score

For a non-terminal position, Ultra evaluates

$$E = S_{\text{count}} + S_{\text{shape}} + S_{\text{reach}} + S_{\text{assign}} + S_{\text{route}} + S_{\text{exposure}}.$$

Every term is measured from the side to move. When the turn changes, the point of view changes with it. This makes the score symmetric between Red and Blue.

Term	Question being asked	Typical scale
Shield count	Who has more surviving cover near the kings?	tens
Shield shape	Whose remaining cover is closer and more useful?	tens
Reachability	How many lasers can still reach each king?	tens
Assignment control	Has a laser become permanently committed to one king?	thousands
Route control	Who can complete the important laser routes sooner?	hundreds to thousands
Exposure	Is a king on a live laser path right now?	hundreds

If the game is already over, these positional terms are ignored:

$$E = \begin{cases} +10000, & \text{the side to move is the winner,} \\ -10000, & \text{the opponent is the winner,} \\ 0, & \text{the result is a draw.} \end{cases}$$

2. King cover

2.1 Shield count

Only shields within a two-square Chebyshev radius of a king count as that king's cover. In other words, a shield counts when it lies no more than two rows and two columns away.

Let N_{own} and N_{opp} be the surviving shield counts around the two kings. Then

$$S_{\text{count}} = 25\,(N_{\text{own}} - N_{\text{opp}})\,.$$

Each extra nearby shield is therefore worth 25 points.

2.2 Shield shape

Distance matters in addition to quantity. A shield one square from its king is worth 18 shape points; a shield two squares away is worth 6. The term is the difference between the two formations:

$$S_{\text{shape}} = Q_{\text{own}} - Q_{\text{opp}}.$$

This distinguishes a compact protective formation from the same number of shields scattered farther away.

3. Future reachability

Imagine that every currently empty legal square may eventually contain either mirror orientation. For each of the two lasers, ask which kings remain reachable in at least one future arrangement.

Let R_{attack} be the number of lasers that can still reach the opposing king, and R_{danger} the number that can still reach the current player's king. Then

$$S_{\text{reach}} = 18\,(R_{\text{attack}} - R_{\text{danger}})\,.$$

This is deliberately a modest term. Merely being reachable is less informative than knowing how difficult the route is or whether the laser has become permanently assigned.

4. Permanent laser assignment

This is the decisive strategic feature introduced in v0.13.4.

The rules require the two lasers to retain compatible future routes to different kings. Suppose one laser can now reach only the Blue king. Because the two target kings must differ, that laser is permanently assigned to Blue and the other laser must serve the route to Red. Later mirror placements can remove options, but they cannot restore a route already made impossible.

For each laser:

$$a_\ell = \begin{cases} +1, & \text{it can reach only the opposing king,} \\ -1, & \text{it can reach only the current player's king,} \\ 0, & \text{it can still reach both kings.} \end{cases}$$

The assignment score is

$$S_{\text{assign}} = 1200 \sum_{\ell=1}^{2} a_\ell.$$

The large weight is intentional. A dedicated attack laser is not a small positional preference: it permanently defines which side can build with that laser and which king must absorb its pressure.

5. Route control

Reachability answers *whether* a route exists. Route control estimates *how many new mirrors* are needed to complete it.

For each laser ℓ, let

$$c_\ell(P) = \min(\text{fewest new mirrors needed to reach king } P,\ 12)\,.$$

An unreachable target is also assigned the capped value 12.

For the opposing king, sort the two costs:

$$\alpha_1 \le \alpha_2.$$

For the current player's king, sort them separately:

$$\delta_1 \le \delta_2.$$

The two race gaps are

$$r = \delta_1 - \alpha_1, \qquad q = \delta_2 - \alpha_2.$$

A positive r means the quickest attack on the opponent is closer than the quickest danger to the current player's king. The reserve gap q compares the second laser and therefore rewards control that is not dependent on a single route.

Define the signed square

$$\sigma(x) = x|x|.$$

It preserves the sign of a lead while making large leads matter more than small ones. For each laser, also define its individual control gap

$$g_\ell = c_\ell(\text{own}) - c_\ell(\text{opponent}).$$

The complete route term is

$$\begin{aligned} S_{\text{route}} &= 67r + 42q \\ &\quad + 28\sigma(r) + 18\sigma(q) + 8\sum_{\ell=1}^{2} \sigma(g_\ell). \end{aligned}$$

Why all three comparisons? The quickest route measures the immediate race, the second route measures resilience, and the per-laser gaps preserve information about which physical laser controls each side of the board.

6. Current laser exposure

The board's present laser paths are evaluated after all existing mirrors:

$$S_{\text{exposure}} = \begin{cases} +540, & \text{only the opposing king is currently hit,} \\ -540, & \text{only the current player's king is currently hit,} \\ 0, & \text{neither king or both kings are hit.} \end{cases}$$

Both kings being hit is neutral in this term because the lasers fire simultaneously. The actual game result, when a move is resolved, still takes priority over every positional score.

7. Reading a score

- A score near zero means the measurable advantages are balanced; it does not guarantee a draw.
- A few hundred points usually indicate route or exposure pressure.
- A swing of roughly 1200 points usually signals that a laser has become permanently dedicated to one king.
- Scores near ± 10000 represent a forced or completed win and dominate every positional consideration.

Example: the permanent-lane failure. If both lasers initially reach both kings, the assignment term is zero. If a move makes the left laser able to reach only Blue, then from Red's point of view the assignment term becomes $+1200$; from Blue's point of view it becomes -1200. This captures the irreversible strategic change even before the final mirror route is built.

8. Worked example

Consider a position evaluated for Red:

- Both sides have four nearby shields, so $S_{\text{count}} = 25(4-4) = 0$.
- Red's shield shape is worth 60 and Blue's is worth 48, so $S_{\text{shape}} = 60 - 48 = 12$.
- Both lasers can reach Blue, while only one can reach Red, so $S_{\text{reach}} = 18(2-1) = 18$.
- One laser can reach only Blue and the other still reaches both kings, so $S_{\text{assign}} = 1200(1+0) = 1200$.
- The sorted attack costs are $(2, 5)$, the sorted danger costs are $(4, 7)$, and each physical laser has a control gap of 2. Consequently,

$$
\begin{aligned}
S_{\text{route}} &= 67(2) + 42(2) + 28\sigma(2) + 18\sigma(2) + 8\big(\sigma(2) + \sigma(2)\big) \\
&= 466.
\end{aligned}
$$

- Neither king is currently exposed, so $S_{\text{exposure}} = 0$.

The complete evaluation is therefore

$$E = 0 + 12 + 18 + 1200 + 466 + 0 = \boxed{1696}.$$

This is a large Red advantage. Most of it comes from the permanent assignment, while the route race confirms that Red is also closer to converting that structural advantage.

This document describes the evaluation used by Laser War v0.13.4.